João André **Roque Costa**

MSc Graduate, Computer Science and Engineering

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AI Developer with a Master's degree in Computer Science (AI and Data Science) from **Instituto Superior Técnico**. Experienced in developing **AI agent solutions**, process automation, and managing **Azure** resource groups. Skilled in Python, Azure Foundry, Azure Blob Storage, MongoDB, and data analysis techniques.

Professional Experience

Deloitte — *Coimbra, Portugal*

Tech Consultant

Apr. 2026 - Oct. 2026

- Developed a multi-agent application for data migration and generation tasks (ClaudeSDK, Azure Foundry, Azure Blob Storage and MongoDB).
- Ported and updated an Azure Resource Group between two subscriptions (Azure Managed Instances, Data Factory, Databricks).
- Automated an HR report generation process using Python and Playright.
- Drafted a project proposal to a client to automate the documentation processes (RAG, OCR).

Instituto Superior Técnico — *Lisbon, Portugal*

Invited Teaching Assistant

Oct. 2024 - Jul. 2025

- Lectured and evaluated practical classes on Databases and Data Science.
- Assisted students with hands-on projects in data treatment, visualization, and analysis techniques using Python (pandas/matplotlib).

Education

Instituto Superior Técnico — *Lisbon, Portugal*

M.Sc. Computer Science and Engineering

Sep 2023 - Nov 2025

- Specialized in **Data Science** with emphasis in **Artificial Intelligence**, maintaining a focus on scalable data architectures and distributed systems.
- Relevant Coursework: Information Visualization, Data and Information Systems Management, Data Analysis and Integration, Machine Learning, Deep Learning, Cloud Computing and Virtualization, Optimization Algorithms.
- Thesis: Maximum Variance Unfolding on Disjoint Manifolds (MVU-DM).
  - Developed a novel extension to the MVU algorithm that enables nonlinear dimensionality reduction on disconnected datasets.
  - Engineered a decomposition-based approach that allows for parallel processing, significantly improving scalability for large-scale datasets that were previously computationally infeasible.
  - Achieved equal-quality embeddings and preservation of local geometric structures compared to the standard method.

Instituto Superior Técnico — *Lisbon, Portugal*

B.Sc. Computer Science and Engineering

Sep 2020 - Jul 2023

- Developed a strong foundation in **Distributed Systems**, **Operating Systems**, and **Database Theory**.
- Collaborated on several full-stack and systems-level projects, emphasizing memory management in C/C++ and relational schema design in SQL.

Extracurricular

Personal Software Projects (Self-Hosted)

- Engineered an automated betting bot for parimutuel systems using a modified Kelly Criterion for risk management and optimal stake sizing.
- Developed a custom Discord-based ELO ranking and automatic matchmaking system.
- Architected a multi-platform arbitrage engine and inventory manager. With automated price monitoring and trade execution across multiple marketplaces.
- Built a full-stack data visualization dashboard to analyze player performance metrics, converting raw API data into actionable strategic insights.

Skills & Competencies

Programming	**Python** (Claude-Agent-SDK, Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Playright), Java, C++, C#
Databases	**SQL**, **NoSQL**, Microsoft SQL Server (SSMS), PostgreSQL, MongoDB, MySQL, pgAdmin
Data Engineering	Databricks, **Pentaho PDI**, SQL Profiler, Tuning Advisor (DTA), DataCleaner tool
Business Intelligence	Pentaho (Server, Schema Workbench, Report Designer), Saiku Analytics
Cloud & Architecture	**AWS**, **Azure**, ArchiMate, Camunda Modeler, Enterprise Architecture (Sparx)
Developer Tools	Git, Linux, Bash Scripting, RESTful APIs, LaTeX
Languages	Portuguese (Native), English (Fluent)